

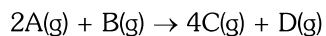
MOLE - CONCEPT

1. Select a correct statement :-
(1) Chemistry studies about matter
(2) Chemistry is a science of atoms & molecules
(3) Both statements are correct
(4) None of them
2. Drugs effective in cancer therapy are :-
(1) Cisplatin (2) Taxol
(3) Both (1) & (2) (4) AZT (Azidothymidine)
3. Characteristics of a liquid are :-
(1) Definite shape & definite volume
(2) Indefinite shape & indefinite volume
(3) Definite shape & indefinite volume
(4) Indefinite shape & definite volume
4. On cooling & heating a liquid it changes its state respectively as follows :
(1) On cooling → Gas
 On heating → Solid
(2) On cooling → Solid
 On heating → Gas
(3) Its can change in either of the states
(4) No change in state will occur.
5. Air is an example of : -
(1) Element
(2) Compound
(3) Homogeneous mixture
(4) Heterogeneous mixture
6. The standard unit for temperature measurement is
(1) Celsius (2) Fahrenheit
(3) Kelvin (4) All of the above
7. Luminous intensity is measured in which unit?
(1) Kelvin (2) Candela
(3) Ampere (4) Mole
8. Length, mass and time are respectively measured in _____ S.I. units?
(1) Centimetre, gram & minute
(2) metre, kg & second
(3) metre, gram & second
(4) centimetre, kg & second
9. Which of the following relations is incorrect?
(1) 1 Kilometre = 1000 metre
(2) 1 picometre = 1000 nano metre
(3) 1 petametre = 10^{30} femto metre
(4) 1 micro metre = 10^{-12} megametre
10. Which of the following relation is incorrect regarding temperature?
(1) $^{\circ}\text{F} = \frac{9}{5}(^{\circ}\text{C}) + 32$ (2) $\text{K} = ^{\circ}\text{C} + 273.15$
(3) $^{\circ}\text{C} = \text{K} + 273.15$ (4) $^{\circ}\text{C} = \frac{5}{9}(^{\circ}\text{F} - 32)$
11. The law stating that a given compound always contains exactly the same proportion of elements by weight was given by :-
(1) Lavoisier (2) Proust
(3) Dalton (4) Avogadro
12. Equal volume of gases contains equal number of molecules at same temp & pressure, this law was given by :-
(1) Berzilius (2) Gay – Lussac
(3) Dalton (4) Avogadro
13. Carbon + oxygen → Carbon dioxide
 12g 32g 44g
Carbon + Oxygen → Carbon monoxide
 12g 16g 28g
This data illustrates which of the following law :-
(1) Law of multiple proportion
(2) Law of reciprocal proportion
(3) Law of conservation of mass
(4) Both (1) & (3)
14. Which statement is incorrect regarding Dalton's Atomic Theory?
(1) Matter consists of indivisible atoms
(2) All atoms of a given element have indential properties
(3) Compounds are formed when atoms of different elements combine in a fixed ratio
(4) Atoms are either created or destroyed in a chemical reaction

- 15.** The term 'amu' stands for :
- Atomic mass unit
 - Unified mass
 - Mass exactly equal to one-twelfth the mass of one carbon - 12 atom
 - All of the above
- 16.** Average atomic mass of elements can be calculated by which of the following formula?
- (M = Molar mass, X = mole fraction, n = moles)
- Avg. atomic mass = $\frac{n_1 M_1 + n_2 M_2}{n_1 + n_2} = \frac{\sum_{i=1}^n n_i M_i}{\sum_{i=1}^n n_i}$
 - Avg. atomic mass = $X_1 M_1 + X_2 M_2 = \sum_{i=1}^n X_i M_i$
 - Avg. atomic mass = $\sum_{i=1}^n (\%age)_i . M_i$
 - All of the above
- 17.** The mass of sodium chloride (NaCl) is known as :
- Atomic mass
 - Molecular mass
 - Formula mass
 - All of the above
- 18.** Which statement is correct regarding mole?
- Mole is one of the seven base quantities in SI system.
 - One mole contains as many particles or entities as there are atoms in exactly 12g of the C-12 isotope
 - The number of entities in one mole is equal to Avogadro number
 - All of the above are correct.
- 19.** Which of the following formulae are correct regarding mole calculations?
- mole = $\frac{\text{mass}}{\text{molar mass}}$
 - mole = $\frac{\text{volume of gas at STP}}{22.4 \text{ litre}}$
 - mole = $\frac{\text{number of entities}}{\text{Avogadro number}(N_A)}$
 - All of the above

- 20.** Which option is correct regarding the percentage composition of CaCO_3 ?
- Ca is 40% by mass
 - O is 48% by mass
 - C is 12% by mass
 - All are correct
- 21.** Pick out the correct statement?
- An empirical formula represents the simplest whole no. ratio of various atoms present in a compound
 - The molecular formula shows the exact number of different types of atoms present in a molecule of a compound
 - Both (1) & (2) are incorrect
 - Both (1) & (2) are correct
- 22.** The empirical formula of a compound is CH_2 & the ratio of molecular mass and empirical formula mass is 2, then pick out the incorrect statement?
- Molecular formula is C_2H_4
 - Molecular mass is 28
 - Empirical formula mass is 14
 - % of carbon in compound is 80
- 23.** Which of the following statement is correct?
- Stoichiometry deals with the calculation of masses of the reactants & the products involved in a chemical reaction
 - According to law of conservation of mass, a balanced chemical equation has the same number of atoms of each element on both sides of equation
 - The reactant which gets fully consumed & limits the amount of product formed is known as limiting reagent
 - All are correct

24. For the following reaction, pick out the incorrect interpretations?



- (1) 2 moles of A & 1 mole of B reacts to form 4 moles of C & 1 mole of D
- (2) 2 gm of A & 1 gm of B reacts to give 4 gm of C & 1 gm of D.
- (3) 2 molecules of A & 1 molecule of B reacts to give 4 molecules of C & 1 molecule of D
- (4) 2 volume of A & 1 volume of B reacts to give 4 volume of C & 1 volume of D (At STP)

25. Which of the following statement is incorrect regarding concentration terms?

- (1) Molarity is the no. of moles of solute present in 1 kg of solvent.
- (2) Molality is the no. of moles of the solute present in 1 kg of solvent
- (3) Molarity is the no. of moles of solute present in 1 litre of the solution
- (4) Mole fraction is the ratio of number of moles of a particular component to the total number of moles of the solution